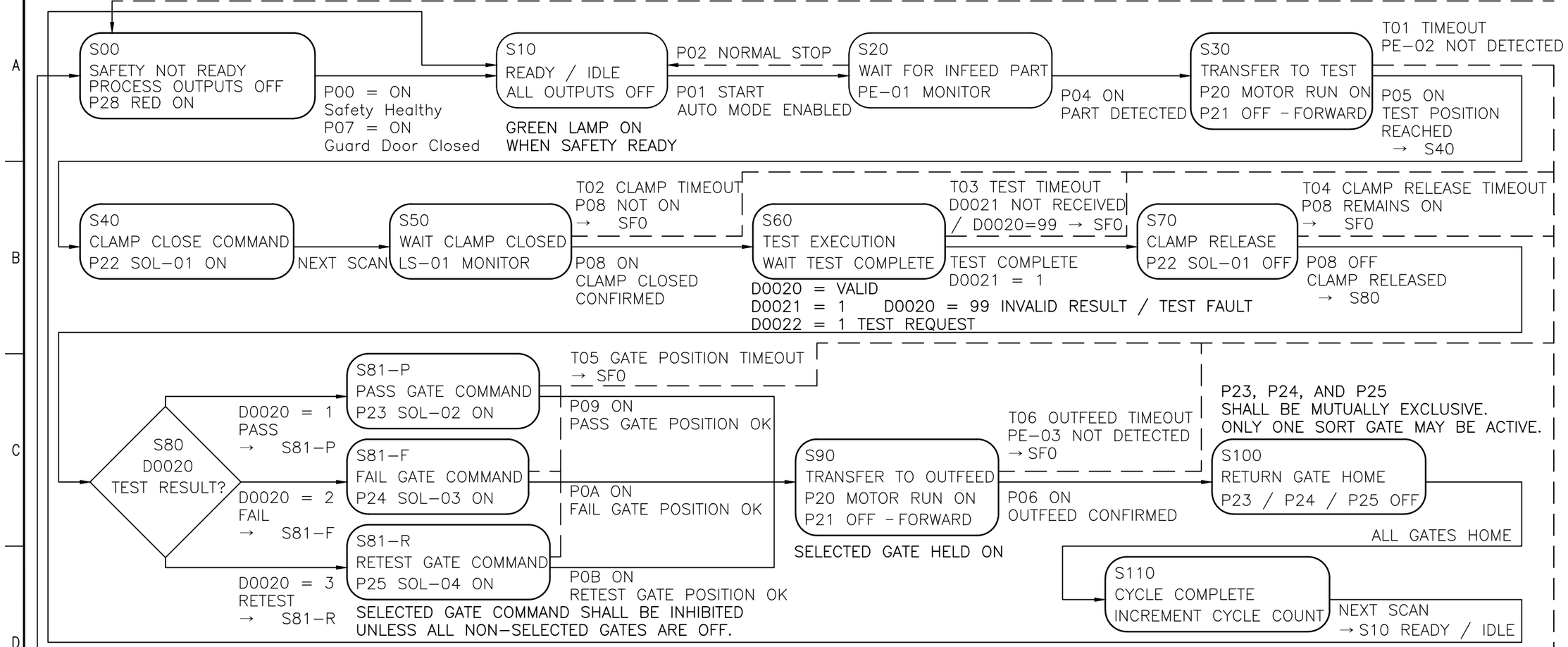
PLC-01 AUTOMATIC CYCLE CONTROL SEQUENCE — CONCEPTUAL PLC STATE DIAGRAM – SEE DWG 004 / 006 / 007 / 008

P00 OFF OR P07 OFF — SAFETY LOSS



STATE LEGEND

State	State Name	Main Action	Primary Feedback
S00	Safety Not Ready	Outputs Off / Initialize	P00, P07
S10	Ready / Idle	Ready for Start	P01
S20	Wait Infeed	Wait for Part	P04
S30	Transfer to Test	P20 ON / P21 OFF	P05
S40	Clamp Close	P22 ON	-
S50	Wait Clamp Closed	Clamp Verify	P08
S60	Test Execute	Wait MCU Result	D0020 / D0021
S70	Clamp Release	P22 OFF	P08 OFF
S80	Result Decision	PASS / FAIL / RETEST	D0020
S81	Gate Command	P23 / P24 / P25	P09 / P0A / P0B
S90	Transfer Outfeed	P20 ON / P21 OFF	P06
S100	Gate Home	P23~P25 OFF	P09/P0A/P0B OFF
S110	Cycle Complete	Update Counts	-
SF0	Fault Hold	Outputs OFF / Alarm	P03 Reset

TEST RESULT INTERFACE DEFINITION

PLC Register	Signal	Value	Definition	Source
D0020	TEST_RESULT_CODE	0	No Result / Test In Progress	MCU-01
D0020	TEST_RESULT_CODE	1	PASS	MCU-01
D0020	TEST_RESULT_CODE	2	FAIL	MCU-01
D0020	TEST_RESULT_CODE	3	RETEST	MCU-01
D0020	TEST_RESULT_CODE	99	Invalid Result / Test Fault	MCU-01
D0021	TEST_COMPLETE_FLAG	0	Test Incomplete	MCU-01
D0021	TEST_COMPLETE_FLAG	1	Test Complete / Result Valid	MCU-01

MCU-01 SHALL WRITE D0020 AND D0021
TO PLC-01 VIA MODBUS RTU RS-485.

HARDWIRED SAFETY CIRCUIT
REMOVES +24V_SAFE_ACT
AND +V_MOTOR_SAFE.
PLC STATE TRANSITION IS MONITORING
/ CONTROL LOGIC ONLY.

NORMAL STOP FUNCTION:
P02 SHALL STOP CONVEYOR MOTION,
REMOVE SOLENOID COMMANDS,
RETURN GATES TO HOME,
AND RETURN PLC TO S10 READY / IDLE.

GENERAL CONTROL NOTES, TIMER SETTINGS,
INTERLOCK MATRIX, AND ALARM MATRIX:
SEE DWG 011 / SHEET 2 OF 2

NO.	DATE	REVISION
A	2026-09-09	INITIAL RELEASE

DWG TITLE
EDS Wafer Map Linked Package
Final Test and Multi-Bin Sorting Cell
PLC CONTROL SEQUENCE

DRAWN BY: J. H. KO
DATE: 2026-09-09
DWG NO: 011
SHEET NO: 1 OF 2